

Data type: A data type defines the kind of data a variable can store and how the computer should interpret it.

Memory size for char = 1 byte whereas double = 8 bytes

Five main memory segments: Text (code), Data, BSS, Heap, Stack.

Data vs BSS: Data stores initialized global/static variables, while BSS stores uninitialized or zero-initialized global/static variables.

Stack and text placement: The stack is placed at high addresses so it can grow downward, while the text segment is placed low so the heap can grow upward with space between them.

Function return: The OS manages the process's overall stack memory.

Forgetting free(): It causes a memory leak, eventually wasting available heap memory.

ptr vs & ptr: ptr is the address stored in the pointer while &ptr is the address of the pointer variable itself.

Why void * with %p: %p expects a void * so casting ensures the pointer has the correct type for printf.

How OS identifies process memory: The OS uses separate virtual address page tables for each process to map and protect its memory.